



Connect a GitHub Repo with AWS



Leah Gibson

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(4/8): git-core-2.50.1-1.amzn2023.0.1.x86_64.rpm           34 MB/s | 4.9 MB   00:00
(5/8): perl-File-Find-1.37-477.amzn2023.0.7.noarch.rpm    460 kB/s | 25 kB   00:00
(6/8): perl-Git-2.50.1-1.amzn2023.0.1.noarch.rpm         756 kB/s | 41 kB   00:00
(7/8): perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64.rpm   1.5 MB/s | 36 kB   00:00
(8/8): perl-lib-0.65-477.amzn2023.0.7.x86_64.rpm         513 kB/s | 15 kB   00:00
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Total                                                    34 MB/s | 7.9 MB   00:00

7/8
Installing       : git-2.50.1-1.amzn2023.0.1.x86_64      8/8
Running scriptlet: git-2.50.1-1.amzn2023.0.1.x86_64      8/8
Verifying        : git-2.50.1-1.amzn2023.0.1.x86_64      1/8
Verifying        : git-core-2.50.1-1.amzn2023.0.1.x86_64 2/8
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perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64  perl-lib-0.65-477.amzn2023.0.7.x86_64

Complete!
[ec2-user@ip-172-31-5-100 ~]$ git --version ←
git version 2.50.1
[ec2-user@ip-172-31-5-100 ~]$ git version 2.50.1
```



Introducing Today's Project!

In this project, I will set up a Git Repository for my web app's code. This is project two of a 7 day devops challenge. By the end of this project, the code that I write for my Java web app will be stored securely in Github.

Key tools and concepts

Services I used were GitHub, Amazon EC2(which was my development instance), key pair, VS Code. Key concepts I learnt include setting up a Git Repository, the difference between Git and GitHub, branches, and the commands for saving and pushing changes into my code.

Project reflection

Not gonna lie this project took me approximately two hours to do including deleting resources. The most challenging part was to learn all the different Git commands. However it was most rewarding to set up my ReadMe and seeing my text reflected on GitHub.

I did this project because I want to grow in my knowledge of CI/CD pipelines. I want to hopefully become a devops engineer and make enough to support myself.

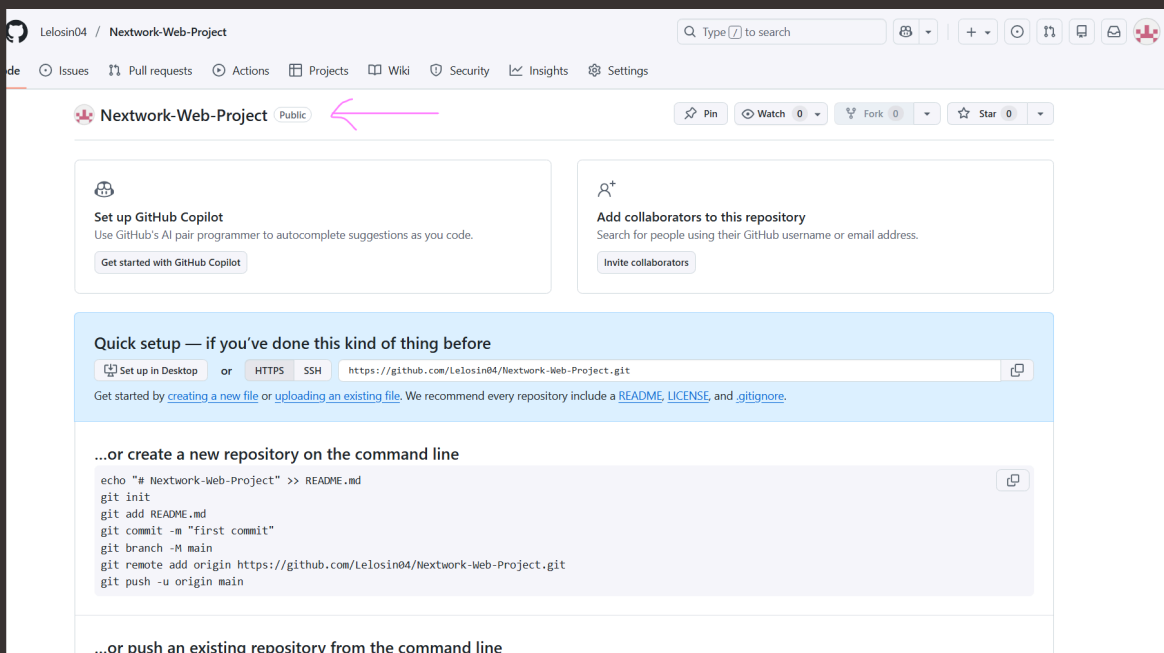
In the next project I will be furthering building this CI/CD pipeline.



Git and GitHub

Git is a version control system which means its used to track changes we make in our code. It is also useful for collaborations. You can see who made what changes and the hand over from development is a loot smooother when everyone;s looking at the same shared repository of code. I installed Git using the commands "sudo dnf update -y" and then "sudo dnf install git -y"

GitHub is a platform that lets us store, share and collaborate on code. It's called GitHub because it uses the tool Git for version control (one of its main features). Github makes collaboration, seeing public repositories, and managing multiple repositories much easier. I'm using GitHub in this project to store my webapp's code.





```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - nextwork-web-project + ▢ ▢ ... ▢ ▢ ▢ ▢

● [ec2-user@ip-172-31-5-100 nextwork-web-project]$ pwd
/home/ec2-user/nextwork-web-project
● [ec2-user@ip-172-31-5-100 nextwork-web-project]$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/ec2-user/nextwork-web-project/.git/
○ [ec2-user@ip-172-31-5-100 nextwork-web-project]$

```



To push local changes to GitHub, I ran three commands

`git add`

The first command I ran was "git add ." which adds our changes to a staging area. It is the place to review all changes made to the code so that I can decide what changes I want to save.

`git commit`

The second command I ran was git commit which is the command for saving the commands in my staging area. Using -m means I am leaving a message for that commit.

`git push`

The third command I ran was "git push -u origin master." This command pushes the code changes I saved to our remote origin. Using '-u' is a special flag that makes our remote origin the default UPSTREAM repository. Next time I have changes I want to push, I can simply run "git push" and Git already knows where to push those changes to.

Authentication

When I commit changes to GitHub, in my case Git asks for my credentials because I was plugging in the wrong url... But normally password authentication was removed in 2021. There are too many security risks associated with entering passwords over the terminal to github. So now better security measures are available.

Local Git identity

Git needs my name and email because its a virtual control system which means it used to track changes to code and who made the changes. To identify who made x chnages to code, git will needs to know your git and email.

Running git log showed me that by default Git is saving our co changes to a username called "EC2 Default User" instead of our actual name and details.

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
bash - nextwork-web-project  + v  [ ]  [ ]  [ ]  [ ]

• [ec2-user@ip-172-31-5-100 nextwork-web-project]$ git push -u origin master
branch 'master' set up to track 'origin/master'.
Everything up-to-date
• [ec2-user@ip-172-31-5-100 nextwork-web-project]$ git log
commit 3890ae37657c42e065fca0f9b38bdc4db9e0dd439 (HEAD -> master, origin/master)
Author: EC2 Default User <ec2-user@ip-172-31-5-100.us-east-2.compute.internal>
Date: Tue Jan 13 19:44:03 2026 +0000

    Updated index.jsp with new content
○ [ec2-user@ip-172-31-5-100 nextwork-web-project]$

```



GitHub tokens

GitHub authentication would normally fail when you enter your password because there are better security measures set in place to gain access to github. Again in my case, I was enetering the wrong URL so it didnt matter what password I was putting. It would'nt have connected even if there was less lenininet security measures set in place.

A GitHub token is a temporary password that grants access to my github account. I'm using one in this project because it lets me safely authenticate to my github repo while in our EC2 instance.

I could set up a GitHub token by first going the developer settings in github. I set one that expires in 7 days and only allows permissions to repos.



Settings / Developer Settings

Q Type [Z] to search

GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

for funsies. I was already let into the club aka ec2 github repo.

What's this token for?

Expiration

📅 7 days (Jan 20, 2026)

The token will expire on the selected date.

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

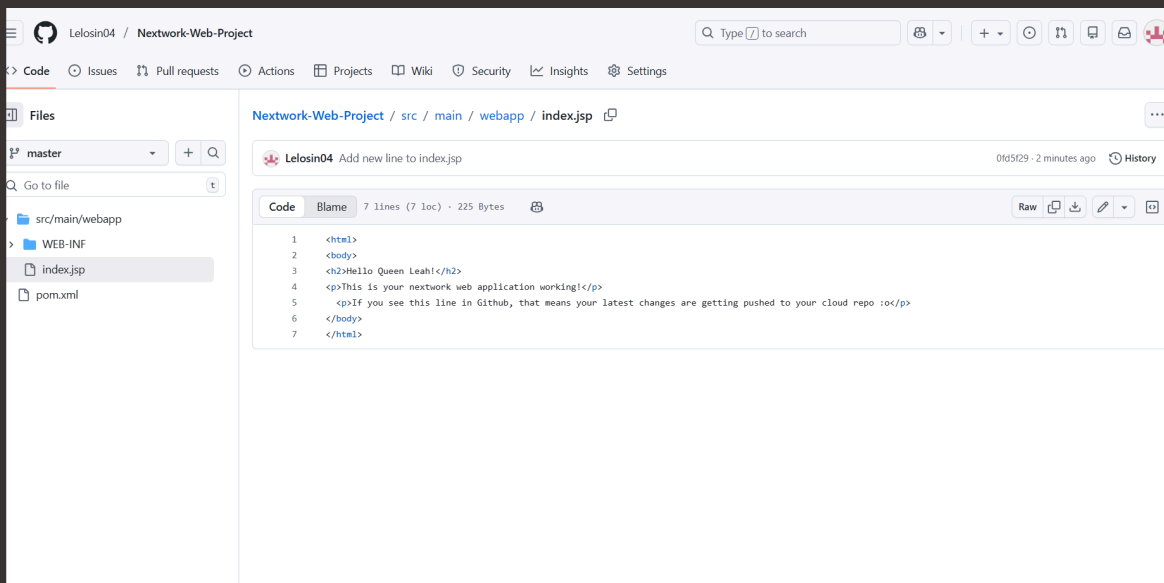
☒ repo	Full control of private repositories
☒ repo:status	Access commit status
☒ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☒ repo:invite	Access repository invitations
☒ security_events	Read and write security events
☐ workflow	Update GitHub Action workflows
☐ write:packages	Upload packages to GitHub Package Registry
☐ read:packages	Download packages from GitHub Package Registry
☐ delete:packages	Delete packages from GitHub Package Registry



Making changes again

I wanted to see Git working in action, so I made another change to my index.jsp file. I couldn't see the changes in my GitHub repo initially because I did not add, commit, or push those changes.

I finally saw the changes to my web app after running those previous commands and then refreshing refreshing my "index.jsp" on my github.



Setting up a README file

As a finishing touch to my GitHub repository, I added a README file, which is a document that looks at the root of my repo and introduces my project, what it does, how to use it, etc. I added a README file by running the command "touch README.md" in my EC2's instance terminal.

My README is written in Markdown because that is the common text language that lets me format text in a document. Special characters can help you format text in Markdown, such as a backtick(') for code or double sterisk(**) for bolding and more.

My README file has 6 sections that outline a table of contents, technological tools i used, an introduction, how to set it up,contact information, and a conclusion.

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